

Breaking the Latency Barrier: 310ms Asynchronous DAG-based BFT Atomic Broadcasts via TEEs

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Byzantine atomic broadcasts enforce consensus on a total order of requests among replicas, forming the foundation of state machine replication (SMR). In recent years, DAG-based architectures have emerged as a powerful solution to overcome the leader-bottleneck inherent in leader-based atomic broadcasts. A DAG-like structure allows all replicas to propose blocks simultaneously, relying on the DAG to naturally resolve any concurrency issues [Ke21]. By avoiding a single, easily overloaded leader and allowing replicas to contribute blocks concurrently, throughput is significantly increased. Besides resolving the leader-bottleneck, running a reliable broadcast on a DAG structure facilitates highly efficient block distribution. Rather than defaulting to redundant network-wide broadcasts, replicas can easily detect and pull missing blocks using certification [Da22; LH23].

Despite their throughput advantages, DAG protocols suffer from higher latency, as ordering a block requires waiting for the arrival of subsequent blocks. While previous research attempted to minimize the number of required block rounds [Ar25; Da22; Sp22], recent work has achieved significantly lower latencies by operating on uncertified DAGs [Ba25; Jo25]. This approach omits explicit block certification during the reliable broadcast, compensating instead by observing structural certification patterns within the DAG. The massive latency improvements gained by eliminating certification directly motivate the use of TEEs. Because a TEE provides communication-free block certification, it allows the system to reach the low latency of an *uncertified DAG* while preserving the advantages of a *certified DAG* [LH23].

In this talk, we explore how TEEs not only optimize the reliable broadcast but also open new avenues for optimizing the ordering process itself. State of the art asynchronous DAG-based protocols on certified DAGs offer an average commit latency between 1150ms and 1380ms [Da22; Sp22], while partially synchronous protocols reach around 900ms [Ar25]. With uncertified DAGs, this latency can be decreased to 500ms [Jo25] and 430ms [Ba25] respectively. Because TEEs allow the system to switch back to working on certified DAGs, we leverage previous research, adapt it to incorporate TEEs, and discuss new ideas to further minimize latency. Ultimately, we reach a latency of 310ms in an asynchronous network. Thereby, we demonstrate that TEEs can close the gap between asynchronous and partially synchronous DAG-based atomic broadcasts, achieving latencies lower than any existing DAG protocol while simultaneously increasing fault tolerance.

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